

## Written Analysis: Train–Test Split Experimentation

This project examined how different data-splitting strategies influence model evaluation for bank fraud detection across balanced (50% fraud) and imbalanced (5% fraud) datasets. Experiments using multiple train-test ratios (60/40, 70/30, 80/20, 90/10), stratified vs regular splitting, and temporal validation demonstrated that splitting methodology significantly affects performance estimates and model reliability.

Across all split ratios, larger training sets slightly improved test accuracy, but also increased the performance gap between train and test sets, indicating mild overfitting. This effect was more pronounced in the imbalanced dataset, where high overall accuracy (up to 95%) largely reflected the dominant non-fraud class rather than genuine predictive capability. These results confirm that accuracy alone is a misleading metric for imbalanced classification tasks such as fraud detection.

Stratified splitting produced markedly more stable evaluations. In the imbalanced dataset, regular random splitting caused large fluctuations in fraud rates across splits, with some test sets containing **0% fraud cases**, making evaluation unreliable. Stratified splitting maintained consistent class proportions, halving accuracy variance and improving ROC-AUC scores by approximately 0.006 across all ratios. This highlights that stratification is essential when fraud prevalence is low.

Temporal validation uncovered a different set of challenges. Random splitting overestimated performance by **1.6%** compared to temporal splitting because it unintentionally leaks future fraud patterns into the training set. Temporal splits revealed genuine performance degradation over time, suggesting concept drift in evolving fraud behaviours. Walk-forward validation provided the most realistic estimate of production performance at **84.0% ± 0.7%**, capturing temporal instability and enabling drift monitoring.

Overall, the experiments show that splitting strategy substantially impacts perceived model performance. For imbalanced datasets, stratification is necessary for fair evaluation, while for time-dependent data, temporal or walk-forward validation provides the most reliable indication of real-world performance. These findings underscore the importance of aligning the validation method with the problem's structure to avoid misleading conclusions and ensure deployment-ready models.